

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(d)	(i)		$n(\text{Al}) = 4.85 \times 10^{-3}$ (1) $n(\text{H}_2) = 7.28 \times 10^{-3}$ (1) volume $\text{H}_2 = 178 \text{ cm}^3$ (1)		3		3	2	
		(ii)		$V = \frac{1 \times 178 \times 323}{298 \times 1.6} (1)$ $V = 121 \text{ cm}^3$ (1) if 200 cm^3 used award (2) for $V = 135 \text{ cm}^3$ Credit alternative method using $pV = nRT$ $V = \frac{7.28 \times 10^{-3} \times 8.31 \times 323}{161600} (1)$ $V = 121 \text{ cm}^3$ (1)		2		2	2	
				Question 10 total	6	8	0	14	8	0